

Storyboard companion

Everything in the project that bears on the step-imagery decision, in one place. Compiled 21 August 2026 from `philosophy.md`, `asset-spec.md`, `authoring-standard.md`, `activities.yaml`, `step-imagery-research.md`, `step-imagery-decision.md`, `recipe-pilot-scheme.md`, and the two authored recipes in the vault.

For: storyboarding a number of recipes by hand, on paper, to settle what goes in the 44 px tile.

Every count here is read from the file, not remembered. Where this document and any older note disagree, re-run `matbakh.py status` and `matbakh.py lexicon` — they are the authority. The one thing here that is *judgement* rather than measurement is the Class M / Class S split in §5, and it is marked as such.

1. What the storyboard is actually deciding

Three of the four layers are closed. Do not spend storyboard time on them.

THE COOK ASKS	ANSWERED AT	CARRIER	STATUS
Where am I?	the arc, 17 px	glyph	closed — <code>philosophy §4.4</code> , nothing else survives 17 px
What do my hands do?	the tile, 44 px	open	← this is the decision
Is it ready?	doneness band	photograph of the state	closed — <code>§5.2</code>
Why? What did I miss?	prose, on request	generated from the object	closed — <code>§3</code> , two views one truth

The three candidates for the tile:

	COST AT 500 RECIPES	SURVIVES 17 PX	RECOLOURS BY STATE	SEPARATES CHOP / MINCE
Static glyph	0	yes	yes	no
Animated glyph	~60–100 h once	yes	yes	probably
Reusable action loop (81-verb library)	~50–60 h once	no	no	yes
Recipe-specific frame (video per recipe)	750–2,000 h	no	no	yes

The four things a storyboard can settle that nothing else can

1. **Does the page grammar survive real recipes at volume?** `philosophy §16.5` says step granularity *"has not yet been tested against real recipes at volume."* A storyboard is that test, minus the cooking.
2. **How many tiles actually land on a page**, and therefore how many glyphs compete for attention in one glance.
3. **Which actions you could not draw** — the honest signal for where a glyph is not carrying its weight.
4. **Whether the ingredient changes the gesture.** This is the one finding that could kill the reusable-library recommendation. See §9.

What a storyboard cannot settle: whether motion helps. You cannot draw motion on paper. That needs the prototype (`prototypes/tile-comparison.html`) or the published page, held next to the paper.

2. The rules a page must obey

From [philosophy §3](#), [§4.1-4.7](#) and [authoring-standard §4.1](#). These are settled — a storyboard page that breaks one is a storyboard of a different product.

Structural

- **One step = one page = one station.** If the cook has to move, that is a new page — even for a single action.
- **A page may hold several actions only if they are order-independent and share one station.** All the knife prep is one page with four tiles. *Sear* and *deglaze* are two pages, because in cooking **the sequence is the information**.
- **Waiting is never a page.** *Simmer 20 minutes* is a timer attached to the preceding action.
- **Target 6–10 pages. 14 is a hard ceiling.** Exceeding it is an editorial signal, not a formatting problem — either the dish is genuinely a project, or the steps are over-atomised. The ceiling is doubly justified: it is about the most that shows as legible thumbnails on one phone screen in the map.
- **Nothing scrolls, anywhere in cook mode.**

The two tests to apply while drawing

- **Granularity:** *a step ends when you next need to look at the screen.* Governed by hands, not grammar.
- **The metric is not page count — it is pages you must re-read.** Turning a page is nearly free; coming back from the stove and reconstructing which half of a page you had already done is expensive.

Carriers — [philosophy §5.1](#)

CARRIER	CONVEYS
Icons	Actions — chop, fold, simmer, rest
Digits + unit symbols	Quantity — 200 g, 20 min, 180 °C
Words	Only the name of the thing, and irreducible judgement

And the architectural consequence, [§5.4](#) : **digits are a live text layer over a static visual grammar, never baked into artwork.** Easy to violate the moment anyone produces composite step graphics — worth holding in mind while drawing, because a hand-drawn storyboard naturally merges them.

3. The anatomy of one page

What each storyboard frame has to account for. Field names are the implemented ones (philosophy §11, reconciled 1 August).

The step — one page

```
- station: stove                # board | stove | bench | grill | oven | serve
  qualifier: { en: takliya, ar: التنقية } # renders "Stove · takliya"
  ordered: true                # only if out-of-sequence ruins the dish
  heat: 3                      # 1 low · 2 medium · 3 high
  tiles: [...]
  timer: { minutes: 40, label: {...}, mass_sensitive: true }
  makes: takliya              # names an intermediate
  photo: assets/....jpg
  doneness: { en: ..., ar: ... }
  note: { kind: never, text: {...} }
```

- **station** is a closed list of six. Multiple pages at one station are normal — molokhia has four **stove** pages, separated by **qualifier**.
- **ordered: true** only if doing the tiles out of sequence ruins the dish. Default **false**. On a single-tile step it is meaningless and warns.
- **photo** and **doneness** travel together. The photograph shows what the words say. Never one without the other.

The tile — one action

```
- do: mince                    # lexicon key — the verb comes free, in 5 locales
  qualifier: { en: no colour, ar: بدون تلوين }
  into: "@soffritto"          # destination — carried in data, NEVER printed
  items:
    - { id: onion, amt: 1, cut: brunoise, cut_mm: 2 }
    - { id: carrots, amt: 60, cut: brunoise, cut_mm: 2 }
  short: hand_mince           # attaches a technique clip
```

or the bespoke path, which is the deliberate escape hatch:

```
- verb: { en: "Pour over, listen for the hiss", ar: اسكب واسمع الشهقة }
  glyph: pour                  # bespoke verb REQUIRES an explicit glyph
  item: "@takliya"
```

Fields: **do** or **verb** · **qualifier** · **item** OR **items[]** · **amt** · **carried** · **cut** · **cut_mm** · **into** · **short** · **glyph** (override).

Two that matter for the storyboard specifically:

- **carried: true** marks the same physical ingredient reappearing at a later station, so the shopping list counts one bird rather than three. On paper, mark carried items differently — it is the single most common validator warning.
- **into: is never captioned**. Settled 19 August: the frame is chosen to show where the food goes, so a caption only repeats the picture. It briefs the imagery; it does not print.

Glyph resolution order — `asset-spec.md`

```
tile.glyph      explicit override on that tile
  ↓ else
activity.glyph  from lexicon/activities.yaml
  ↓ else
ingredient.glyph from ref/ingredients.yaml
```

The activity glyph wins over the ingredient glyph. This was a real bug fixed on 19 August — the builder was drawing the *ingredient*, which inverts `philosophy §5.1` and the asset-spec rule "*draw the action, not the ingredient*." It matters on paper too: ingredient glyphs are coarse categories (45 ingredients share `jar`, 38 share `vegetable`), so a page of four spices drew four identical jars and the 17 px arc degraded with it.

4. True geometry — draw your frames at these proportions

From `asset-spec.md`, measured from the prototypes rather than chosen.

ELEMENT	RENDERED	NOTES
Cook-mode tile glyph	44 × 44 px	the primary size, and the minimum touch target
Tablet tile glyph	62 × 62 px	largest use
Arc thumbnail glyph	~17 px	40 × 38 box at <code>scale(.42)</code> . The binding constraint
Inline chrome	32 / 24 / 22 px	
Minimum touch target	48 px	nothing tappable smaller — wet hands
Hero photo	393 × 318, <code>contain</code>	shoot 5:4 landscape (1.25 vs the band's 1.236)
Step / doneness photo	361 × 176, <code>cover</code>	2.05:1 — shoot 4:3 or 3:2, subject in the middle 65% vertically
Arc thumbnail photo	~54 × 42	1.29:1
Shorts poster	84 × 64	1.31:1

Type scale — `tokens.css`: quantity digits 44 px · the action 22 px · the noun 17 px · body 15 px · mono labels 11 px. Nothing functional below 15 px; cook mode is read at arm's length.

An icon must survive a 3.6× range, 17 px to 62 px. An icon legible at 17 px is legible everywhere; the reverse is not true. Drawing rules: **three strokes or fewer**, no detail below 2 px on the 24 grid, **silhouette must differ from every neighbour** — test by squinting at the arc strip.

5. The vocabulary — all 81, by station

Read from `content/lexicon/activities.yaml`. 81 activities, 78 distinct glyphs, 3 shared-glyph pairs.

Class M / Class S is my judgement, not a measurement — M means a single frame cannot separate it from at least one neighbour; S means a placement, a configuration, or an elapsed condition that a still captures entirely. Correct it as you draw. Totals: **47 M / 34 S**.

board — 10 (10 M / 0 S)

KEY	CLASS	GLYPH
chop	M	chop  shares with mince
crush	M	crush
cut_wedges	M	wedges
dice	M	grid-dots
grate	M	grater
mince	M	chop  shares with chop
peel	M	peeler
shred	M	line-dashed
slice	M	slice
zest	M	lemon

The board is 10 for 10 motion. This is not a coincidence — it is why `chop` and `mince` share a file.

stove — 19 (15 M / 4 S)

KEY	CLASS	GLYPH
bain_marie	S	bowl-over-pot
blanch	M	bubbles-fast
boil	M	bubbles
braise	S	pot-deep
deep_fry	M	deep-fry
deglaze	M	deglaze
do_not_stir	S	nostir
flambe	M	flame-pan
fry	M	fry
ladle	M	ladle
poach	M	egg
pressure_cook	S	pressure-cooker
reduce	M	arrow-down-right
simmer	M	pot
skim	M	spoon
steam	M	steam
stir	M	rotate
stir_fry	M	wok

KEY	CLASS	GLYPH
sweat	M	steam-low

boil, simmer and poach are defined by bubble behaviour. As stills they are three pans of water.

bench — 34 (14 M / 20 S)

KEY	CLASS	GLYPH
blend	M	blender
brine	S	bowl-salt
brush	M	brush Δ shares with baste
chill	S	snowflake
coat	S	coat
cool	S	temp-down
cure	S	salt-cure
drain	S	strain
dry	S	sun
ferment	S	ferment
flatten	M	flatten
fold	M	fold
freeze	S	ice
grind	M	mill
infuse	S	infuse
inject	S	syringe
knead	M	hand-grab
layer	S	layers
marinate	S	bowl-spoon
mash	M	mash
mix	M	mix
rest	S	clock-pause
roll	M	roll
set	S	set
shape	S	shape
sift	M	sieve
soak	S	bowl
squeeze	M	squeeze
strip	M	strip
stuff	S	stuff
toss	M	toss
wash	S	droplet
whisk	M	whisk
wrap	S	wrap

The bench is where Class S concentrates — 20 of 34. A loop of rest is four seconds of nothing happening.

grill — 4 (3 M / 1 S) · oven — 5 (1 M / 4 S) · serve — 6 (3 M / 3 S) · any — 3 (1 M / 2 S)

KEY	STATION	CLASS	GLYPH
char	grill	M	flame
grill	grill	S	grill
sear	grill	M	sear
skewer	grill	M	skewer
bake	oven	S	bread
baste	oven	M	brush Δ shares with brush
broil	oven	S	flame-top
roast	oven	S	oven
toast	oven	S	toast
drizzle	serve	M	drizzle
dust	serve	M	dust
garnish	serve	S	leaf
serve	serve	S	plate
sprinkle	serve	M	sprinkle
to_taste	serve	S	salt Δ shares with season
add	any	S	add
pour	any	M	pour
season	any	S	salt Δ shares with to_taste

The three glyph collisions, and which imagery fixes them

PAIR	CLASS	DOES MOTION FIX IT?
chop / mince	M / M	Yes — different motions
brush / baste	M / M	Yes — different motions
season / to_taste	S / S	No. Same physical act; what differs is <i>intent</i>

season and to_taste both appear in molokhia. No photograph, frame or loop can show intent — that one is a lexicon or layout problem and survives every imagery option. **Storyboard it deliberately and see whether the page can separate them.**

(A correction worth carrying: earlier notes list cut_wedges / zest as a collision pair. On disk they carry distinct glyphs — wedges and lemon. Three pairs share a glyph, not four. Sole meunière and sayadeya were chosen partly to test that pair, so it is still worth watching in practice, but it is not a shared-file collision.)

The seven demotions — activity + qualifier, no icon, no new word

Recipes referring to the old keys must use the form on the right.

OLD KEY	BECOMES
joint	slice + into pieces / لقطع
simmer_covered	simmer + covered / مغطى
add_to	add + into / في
pour_over	pour + over it / فوقها
arrange	serve + arranged / مرصوف

OLD KEY	BECOMES
alongside	serve + <i>alongside</i> / بجانبها
pickle	ferment + <i>pickled</i> / مخلل

do_not_stir, to_taste and serve were demoted in review but **kept** — they have no underlying activity to qualify. do_not_stir in particular *inverts* its verb: rendered as stir with a qualifier it would read as "**stir**" on a 44 px tile, on the one step where stirring splits the dish.

The escape hatch

A bespoke verb: is allowed and is **working as designed** — recipe 1 used it twice. It requires an explicit glyph: . Log every use: a verb reached for in ≥ 3 recipes earns a place in the closed set; reached once or twice, it stays bespoke. *Do not grow the vocabulary for a long tail.*

6. The four other closed vocabularies

Stations (6): board · stove · bench · grill · oven · serve

Note kinds (4) — free-text headers would destroy the convention a cook learns to read:

KIND	SEVERITY	READS AS	USE FOR
coming_up	info	COMING UP	something to prepare for two pages ahead
stay_here	warn	STAY HERE	do not walk away from the pan
never	danger	NEVER	an absolute — <i>never let it boil, a boil splits it</i>
make_ahead	info	MAKE AHEAD	can be done the day before

Spend **never carefully. Its power comes from being rare.** It is also the one piece of prose that speaks unasked — everything else sits behind an ①.

Units (7): g · kg · ml · L · tsp (rounds to ¼) · tbsp (rounds to ½) · count

Scaling classes (4):

CLASS	BEHAVIOUR	TYPICAL
continuous	linear, rounded to a legible increment	flour, rice, stock, meat by weight
discrete	linear, then whole or ½ if divisible	eggs, lemons, onions
seasoning	sub-linear — factor^{0.75}	salt, chilli, spices
fixed	does not scale	frying oil, blanching water

Tripling a stew and tripling its cumin produces an inedible stew. And nobody wants 750 ml of frying oil because they doubled the koftas — oil depth is set by the pan, not the batch.

Locales (5): en · ar · ar_eg · ar_lv · ar_gulf, falling back ar_gulf → ar_eg → ar. **Dialects are lexicon-only;** per-recipe prose stays in ar.

7. Two worked storyboards, as authored

The only two recipes that exist. Draw against these — they are the calibration.

Molokhia bil farakh — 8 pages, 14 activities, 4 demotions

STATION · # QUALIFIER	HEAT	TILES	TIMER	PHOTO + DONENESS
1 board	—	chop onion ×2 <i>fine</i> · crush garlic ×8 · chop coriander · slice chicken into pieces ← demotion	—	✓ Onion in even dice — no piece thicker than a matchstick
2 stove · stock	2	simmer covered ← demotion (chicken carried, water 1500) · add bay leaf	40 min	✓ Meat lets go of the bone when you push it
3 bench	—	strip chicken carried · drain @stock 1200	—	— · note coming_up
4 stove · takliya	3	fry (garlic carried, ghee) · bespoke "Then add" glyph spice	—	✓ Straw-gold at the edges. One shade past this is bitter · note stay_here
5 stove · pot	1	add into the stock molokhia 400 → @stock	8 min ← mass-sensitive	✓ Thick enough to coat the spoon. No bubbles at the surface · note never
6 stove · pot	1	bespoke "Pour over, listen for the hiss" glyph pour · do_not_stir after	—	—
7 grill	3	sear chicken carried skin-side · season salt	6 min ← mass-sensitive	✓ Lacquered, blistered in places, dry to the touch
8 serve	—	serve alongside rice ← demotion · cut_wedges lemon · to_taste salt	—	✓ Soup, rice and lemon on the table together

What recipe 1 already told us. 14 activities and 4 demotions, 18 coverage cells. Every demotion read naturally as activity + qualifier. Only two bespoke verbs — "Pour over, listen for the hiss" (the escape hatch working as designed, **keep**) and "Then add" (which should simply be **add** — **lint, no lexicon change**). Nothing missing; no collision hit in practice.

Note page 8: **season** (page 7) and **to_taste** (page 8) both occur. They share **salt.svg**. Different pages, so it did not bite — but that is luck, not design.

Tagliatelle al ragu — 9 pages, the stove arc

STATION · # QUALIFIER	HEAT	TILES	TIMER	PHOTO + DONENESS
1 board	—	mince → @soffritto — onion 1, carrots 60 g, celery 60 g, all cut: brunoise, cut_mm: 2	—	✓ Grains the size of a match head, cut not crushed
2 stove · soffritto	1	sweat no colour (3 carried + olive oil) · makes: soffritto	12 min ← mass-sensitive	✓ Slumped and glassy, no gold at the edges
3 stove · meat ordered	3	fry pancetta until the fat runs → sear beef	—	✓ Browned in patches and still sizzling · note never
4 stove · pot ordered	2	add @soffritto to the meat · deglaze wine	6 min	✓ The pan quiet and the base dry again
5 stove · pot	1	add (tomatoes, paste) · season salt · makes: ragu	—	✓ Barely more than a slick of red through the meat
6 stove · covered	1	simmer covered @ragu · add stock a ladle when it tightens	90 min	✓ One lazy bubble every few seconds at the edge · note stay_here
7 stove · uncovered	1	add milk → @ragu	90 min	✓ Dark orange, glossy, thick enough to leave a channel
8 stove · pasta ordered	3	boil (water, salt carried) · add tagliatelle · drain keep a cup of the water	3 min ← mass-sensitive	✓ Floats and the edge has lost its chalk

STATION ·					
#	QUALIFIER	HEAT	TILES	TIMER	PHOTO + DONENESS
9	serve ordered	—	toss in the pan, off the heat (pasta carried + @ragu) · grate parmesan over	—	✓ Every ribbon coated and nothing pooling on the plate

What this one demonstrates: one tile carrying **three ingredients with individual amounts and cuts** (page 1), the **into:** destination, two intermediates (**@soffritto**, **@ragu**), six of nine pages at the **same station** separated only by **qualifier**, and 180 minutes of passive time that produce **no pages at all** — waiting is a timer, never a step.

Both recipes are 8–9 pages, inside the 6–10 target. That is a sample of two. Whether it holds is exactly what the storyboard tests.

8. Which recipes to storyboard

The pilot's fifteen were chosen for *coverage*, not convenience (`recipe-pilot-scheme.md §1`). Storyboarding is far cheaper than cooking, so you can do more of them — but if you want the smallest set that stresses the imagery question specifically, the coverage matrix points at six:

#	DISH	WHY, FOR IMAGERY
1	Molokhia	Already authored — the control. Carries <code>season / to_taste</code> , the collision no imagery fixes
2	Bolognese	Already authored — the stove arc, <code>boil / simmer</code> as motion, multi-item tiles
4	Sole meunière	<code>baste ↔ brush</code> collision and the lemon pair in one dish
6	Lamb kofta	<code>mince ↔ chop</code> collision, plus the only carrier of the grill station (<code>grill</code> , <code>char</code> , <code>skewer</code>)
13	Baladi bread	The Class S extreme — <code>ferment</code> , <code>rest</code> , <code>set</code> , <code>prove</code> . Where motion shows nothing
2b	Koshari	Assembly and layering, the <code>arrange</code> → <code>serve</code> +qualifier demotion, an unusually flat page shape

Two are already authored, so that is four new. If you want a seventh, **mahshi** adds a sub-recipe (`uses / yield`), which changes page shape in a way nothing else in the list does.

A discipline worth borrowing from the pilot: two waves, not one. Storyboard a first batch, log everything, **change nothing**, then do the rest and re-audit. Deciding after the first batch over-fits to it.

9. What to record while drawing

A storyboard that produces an impression settles nothing. A storyboard that produces a dataset settles it. Per tile:

COLUMN	VALUES	WHY
Carrier verdict	glyph sufficient / needed motion / needed <i>this specific dish</i>	The three-way version of the question. ~150 judgements across 15 recipes — the only dataset on this that will ever exist
Ingredient changed the gesture?	y / n	The falsifiable one. ≥3 recipes = the action is not generic and the reusable library fails
Could you draw it in ≤3 strokes?	y / n	The asset-spec rule, tested by hand
Silhouette clash on this page	which pair	asset-spec.md : squint at the arc — if two steps look alike, the cook loses their place
Reached outside the lexicon	the bespoke wording	Feeds the ≥3 rule
Demotion felt forced	y / n	≥2 recipes promotes it back
Arabic felt wrong	the word	One real context is enough to fix a cell

Per page: station · tile count · **ordered** y/n · timer y/n · has doneness y/n · would you re-read this page.

Per recipe: page count against 6–10/14 · activities used · bespoke count · demotions used · sub-recipes · **minutes to storyboard it.**

The tracker already has the shape of this — 03-catalogue/recipe-pilot-tracker.xlsx, Recipes / Coverage / Lexicon-decisions tabs. *Clear the stale Excel lock file ~\$matbakh-lexicon-review-annotated.xlsx in 03-catalogue/ref/ before the verdicts land.*

The thresholds, applied only after the batch

- **Add a verb** to the closed set only if reached for in ≥3 recipes.
- **Promote a demotion back** if its activity+qualifier read awkwardly in ≥2.
- **Split or redesign an icon** if a glyph failed at 17 px anywhere, or two actions collided on one picture in practice.
- **Fix an Arabic cell** if a drafted verb felt wrong in even **one** real context.
- **An unexercised activity means *examine*, not *delete*.**

Exit criteria — all four must hold before freezing the lexicon

1. ≥90% of actions mapped to an existing activity or a clean activity + qualifier.
2. No new verb recurred ≥3× without being addressed.
3. Every icon used read at **both 44 px and 17 px**, no unresolved in-practice collision.
4. Every drafted-Arabic verb that appeared was confirmed or corrected.

10. Traps

Things that will bite while you draw, each learned the hard way.

- **ordered: true on a single-tile page is meaningless** — the one real nit found in recipe 1.
- **A missing carried: true is the most common warning** — *"X already bought in step N"*. On paper, mark carried items distinctly or the basket triples the chicken.
- **Six of the 81 activities currently draw the ingredient, not the action** — `cut_wedges`, `zest`, `garnish`, `season`, `to_taste`, `simmer`. This violates asset-spec's own rule. If your storyboard reproduces it, that is the placeholder glyph misleading you, not the design.
- **Ingredient glyphs are coarse**. 45 ingredients share `jar`, 38 share `vegetable`. A page of four spices is four identical pictures.
- **The activity glyph must win over the ingredient glyph**. Fixed in the builder 19 August; easy to invert again by hand.
- **into: is data, not a caption**. Never write it on the frame.
- **Digits are a live layer**. Do not draw a quantity into the picture — scaling, unit toggles and live pricing all change it.
- **Waiting is never a page**. Bolognese has 180 passive minutes and zero pages for them.
- **The 81 glyph names are not all verified against tabler.io — several are invented** (C-05, still open). The 20 SVGs in `design/icons/` plus 8 cut glyphs are **geometric placeholders**, and the prototype renders them at full fidelity, so a reviewer reads them as the icon set. They are not.
- **Never hand-type a measurable number**. The ingredient count has been quoted as 177, 178 and 179 inside one week. `matbakh.py status` generates them.
- **check counts the repo fixture as a recipe** — `content/recipes/` holds a second molokhia so a bare clone validates. A raw `check` reads *3 authored* where two are real; `status` splits them.

11. The evidence, in one place

For weighing what you see on paper against what is known.

Settled by Matbakh's own documents

- Photographs cannot serve the 17 px arc — `philosophy §4.4`, `asset-spec.md`. **The icon set gets built under every option.**
- Doneness belongs to photography — `§5.2`. Irreducibly per-recipe.
- Prose is two renderings of one object, never authored separately — `§3`.
- An asset that carries no words is made once — `asset-spec.md`, the founding principle, and the argument the reusable library extends.

Strong external evidence

- **Animation over static pictures for procedural-motor knowledge: $d = 1.06$** (95% CI 0.72–1.40), the largest moderator in Höffler & Leutner's 26-study meta-analysis. Realistic/video-based: $d = 0.76$. Overall: $d = 0.37$.
- **But length reverses it** — Wong et al. on the transient information effect: short animations win, long ones lose the advantage entirely.
- **Coherence beats multimedia** — Mayer's coherence principle runs at $d = 1.66$, *larger* than the multimedia principle. Extraneous detail hurts. A `chop` glyph is the photograph with the coherence violation removed.
- **A still is bad at action, good at state** — Heiser et al. A frozen frame mid-chop cannot distinguish chop from slice from mince.
- **Glance cost is set by encoding, not size** — a 10× spread on the same tiny screen from encoding alone.
- **Abstraction, not photography, is what fails cross-culturally** — concrete symbols 75–87.5%, abstract 0–12.5%. And the sobering Egypt-specific one: sequential image comprehension is a **learned convention**, not a perceptual universal.

Precedent

- Every adjacent domain shoots the vocabulary once: Fitbod 1,600 exercises, Physiotech 20,000+, Spreadthesign 610,000 clips across 35 sign languages. Libraries trade as commodities at \$0.09–\$0.20 per clip.
- **No cooking product composes an action-keyed library into recipe steps.** Clean negative, searched by product, capability, patent and academic literature. Nobody has done it because it needs a closed verb vocabulary to key against — which is exactly what Matbakh has and nobody else does.
- **EPIC-KITCHENS-100** annotates unconstrained real kitchen footage with **97 verb classes**. Independent corroboration that the vocabulary closes near 81.

What nobody knows

There is **no published data comparing recipes with step photographs against recipes without**, and **no controlled study of pictorial versus photographic instruction under kitchen conditions at all**. This decision cannot be settled by citation. It can be argued from craft, or measured in-house — which is what the storyboard is.

12. Where everything lives

Design decisions	<code>matbakh-design/design/philosophy.md</code>
Asset geometry and craft	<code>matbakh-design/design/asset-spec.md</code>
How to author a recipe	<code>matbakh-design/design/authoring-standard.md</code>
The imagery research	<code>design/step-imagery-research.md</code> (18 Aug)
The imagery decision	<code>design/step-imagery-decision.md</code> (21 Aug)
The live comparison	<code>prototypes/tile-comparison.html</code>
The vocabulary	<code>content/lexicon/activities.yaml</code>
The pilot scheme	<code>matbakh-private/03-catalogue/recipe-pilot-scheme.md</code>
The tracker	<code>matbakh-private/03-catalogue/recipe-pilot-tracker.xlsx</code>
The two authored recipes	<code>matbakh-private/03-catalogue/recipes/{molokhia,bolognese}.yaml</code>
Refresh the measured numbers	<code>cd content && python3 matbakh.py status --write ../../matbakh-private/02-strategy/matbakh_pm_log.md</code>

Reading order, if you only read three things before starting: `authoring-standard.md` §4.1 (decide the pages before you type) · `philosophy` §4.1–4.4 (page structure, re-reading, tile state, the map) · §5 and §7 of this document (the vocabulary, and the two worked storyboards).

Per tile — the carrier verdict

One row per tile. **The second column is the one that decides it** — if a generic loop would mislead because *this* ingredient changes the gesture, tick it. Three recipes showing that and the reusable-library recommendation fails. Thresholds apply **after** the batch, never during.

RECIPE	PG	ACTIVITY	CARRIER VERDICT GLYPH · MOTION · THIS DISH	INGREDIENT CHANGED THE GESTURE? Y · N	DRAWABLE IN ≤3 STROKES? Y · N	SILHOUETTE CLASH WHICH PAIR	REACHED OUTSIDE THE LEXICON THE BESPOKE WORDING	DEMOTION FORCED? Y · N	ARABIC FELT WRONG THE WORD
			☐ ☐ ☐	☐ ☐	☐ ☐			☐ ☐	
			☐ ☐ ☐	☐ ☐	☐ ☐			☐ ☐	
			☐ ☐ ☐	☐ ☐	☐ ☐			☐ ☐	
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			☐ ☐ ☐	☐ ☐	☐ ☐			☐ ☐	

Carrier verdict: ① glyph sufficient ② needed motion ③ needed this specific dish

Add a verb only if reached for in ≥3 recipes · promote a demotion back if it read awkwardly in ≥2

Per page — did the grammar hold?

One row per page you draw. **The last tick is the real metric** — philosophy §4.2: the expensive thing is not turning a page, it is coming back from the stove and reconstructing which half you had already done. Page count is the wrong thing to optimise; pages you must re-read is the right one.

RECIPE	PAGE	STATION · QUALIFIER	TILES COUNT	ORDERED Y · N	TIMER Y · N	HAS DONENESS Y · N	WOULD YOU RE-READ THIS PAGE?	NOTES — WHAT MADE IT HARD TO DRAW
				☐ ☐	☐ ☐	☐ ☐	☐ ☐	
				☐ ☐	☐ ☐	☐ ☐	☐ ☐	
				☐ ☐	☐ ☐	☐ ☐	☐ ☐	
				☐ ☐	☐ ☐	☐ ☐	☐ ☐	
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				☐ ☐	☐ ☐	☐ ☐	☐ ☐	

One step = one page = one station. Several actions only if order-independent and one station. Waiting is never a page. ordered: true only if out-of-sequence ruins the dish — and never on a single tile

Per recipe – the summary

One row per recipe. Both authored recipes landed at **8–9 pages** — molokhia 8, bolognese 9. That is the calibration, and it is a sample of two. Two waves, not one: draw a first batch, log everything, **change nothing**, then draw the rest and re-audit.

[illegible]

Exit criteria: ≥90% of actions mapped · no new verb recurring ≥3× unaddressed · every icon read at both 44px and 17px · every drafted-Arabic verb confirmed

Log into **recipe-pilot-tracker.xlsx** afterwards